

Define Problem Statements

This document uses the Customer Problem Statement Template to articulate the core challenges faced by each stakeholder group targeted by OptiCrop.

Problem Statement Template

Format: I am [customer] trying to [goal] but [barrier] because [root cause] which makes me feel [emotion].

PS-1 Smallholder Farmer

I am	A smallholder farmer in rural Maharashtra with 2 acres of land
Trying to	Select the most profitable and suitable crop for the next planting season
But	I don't know which crop will grow best given my soil's nutrient levels and the local rainfall forecast
Because	I have no access to agronomic advisory services or soil testing laboratories nearby
Which makes me feel	Anxious, uncertain, and financially vulnerable a wrong choice means a failed harvest and debt

PS-2 Agricultural Researcher

I am	An agricultural researcher at a state university
Trying to	Identify correlations between soil parameters and crop yield across multiple districts
But	Analysing raw datasets manually is time-consuming and error-prone
Because	Current tools lack automated ML pipelines that handle multi-feature agricultural data
Which makes me feel	Frustrated by slow research cycles and the inability to publish timely policy recommendations

PS-3 Agribusiness Company

I am	A procurement manager at an agribusiness company
Trying to	Predict regional crop availability three months ahead for supply chain planning
But	Crop pattern data from farmers is fragmented and unreliable
Because	There is no centralised data-driven platform that aggregates soil and climate signals

Which makes me feel	At risk of supply shortfalls and revenue loss due to poor forecasting
---------------------	---

PS-4 Government Policymaker

I am	A district agricultural officer responsible for food security planning
Trying to	Advise farmers on crop diversification to reduce mono-crop dependency
But	Policy recommendations are based on last year's data rather than real-time environmental signals
Because	No tool exists that combines live soil-nutrient data with weather patterns for regional advisory
Which makes me feel	Ineffective and concerned that my guidance may be outdated by planting season

Root Cause Summary

All four problem statements share a common root cause: the absence of an accessible, data-driven decision support system that translates raw soil and climate data into actionable crop recommendations. OptiCrop directly addresses this gap.